

IN1901

**Microcontroller Based Application Development
Project**

Final Report

PADELPRO



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Index Number

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1.Introduction

In the recent past, there has been significant impact of microcontroller and robotics technology on interactive games. Traditional sports, although highly interactive, usually require physical presence and are normally limited by physical constraints of players or environmental conditions. PadelPro offers an engaging robotic gaming system, which combines the excitement of traditional soccer with the precision of robotic manipulation and control as well as the strategy involved. PadelPro allows players to remotely operate robotic cars that have been created to play competitively, instead of competing physically, allowing players of all ages and backgrounds to experience the excitement of soccer despite body limitations.

The PadelPro game is designed as a competitive two-player game, whereby each competitor operates a specialized robotic car remotely in a designated court. The court is divided into two halves with goalposts in either end similar to those used in traditional soccer. The primary objective of the game is to maneuver the robot car properly to kick/pass the ball into the opposing player's goal through a push-pull type linear solenoid system. Each of the robotic cars is properly equipped with an Arduino Uno microcontroller that governs the actions of the car and a solenoid-operated paddle to kick the ball.

The game also includes two remote controllers, one for each player, with an Arduino Nano microcontroller, and thumb joystick modules which give players exact real-time control of their robotic cars. Controls for movement are intuitive, with separate joysticks for forward/backward movements and left/right movements, while the simultaneous press enables kicking actions through the solenoid. Scoring is monitored with the use of a manual scoreboard that has an LCD display, score increment/decrement buttons, and excellent feedback to encourage competitive spirit. RFID reset is also leveraged when restarting games.

1.1 Problem in Brief

Gaming experiences have evolved considerably over the past few years from mere digital form to sophisticated and rich physical experiences. Even so, there remains a wide gap to be bridged in bringing robotics into competitive, interactive gaming that is just as accessible and enjoyable to all types of players, including those who may be limited by physical hurdles or otherwise prefer non-contact play situations. Traditional physical sports require a lot of physical exertion, which might not be possible or preferable for all and thus limits participation and inclusivity.

Secondly, existing robotic games are seen to focus on robotic precision tasks and not the synergistic mix of the strategic and dynamic aspects of popular sports. Therefore, there exists a real need for a new gaming experience that integrates robotic control and traditional sports mechanics and thus offers an inclusive, participative, and competitive experience to a broader base.

1.2 Significance of Study

The PadelPro project is extremely useful in academic and recreational environments. Through the convergence of robotics engineering and sports dynamics, the project offers a unique environment through which contestants can engage in strategic, skill-based competitions unhampered by physical limitations. PadelPro allows players of all types an opportunity to refine key skills such as precise control, tactical thinking, and solving problems in real time in a challenging, competitive atmosphere.

Moreover, the project contributes importantly to the new field of interactive robotics with demonstrations of wireless communication, microcontrollers, sensor fusion, and actuator control in game-based entertainment. It is an engaging learning tool that inspires learners towards the STEM field by bringing out the union of hardware and software engineering in applied manners.

Furthermore, PadelPro opens access and inclusion, allowing players who may otherwise be unable to or not prefer to compete in sport activity as per traditional sport, to engage in a thrilling and engaging sport without physical exertion. Thus, research and application of PadelPro are a significant advancement in interactive robotic gaming application, fostering innovation and bringing ability for more people to participate in the gaming world.

1.3 Aim and Objectives

Aim:

Develop a dual-player interactive robotic soccer game controlled by Bluetooth joysticks, using Arduino – based robots equipped with solenoid-based striking mechanism, providing a competitive and engaging gaming experience.

Objectives:

- To design and build two robotic platforms controlled by Arduino microcontrollers and utilizing L293D motor driver shield to control movement.
- To implement Bluetooth-based wireless joystick control.
- To integrate solenoid based striking mechanism on each robot to strike the ball to opponent's goal.
- To develop a manual scoring mechanism to track points of each player in a game.
- To create a competitive gaming environment that mimics soccer and tests the player's skill, strategy and timing.

2. Literature Study

Interactive robotic gameplay has seen significant interest and improvement in the recent years, especially in educational and hobbyist domains. Arduino platforms are adopted widely due to its open-source ecosystem, low cost and ease of interfacing with wireless modules and actuators. Bluetooth-based systems using HC-05 module have been applied in several studies to transmit analog joystick inputs between Arduino boards for wireless robot control [1]. However, potential Bluetooth latency challenges have also been noted, indicating that code refinement and signal controlling is crucial for efficient joystick control.

Lui et al. [2] developed a high-speed air hockey game which was a significant study on reactive robotic gameplay, which emphasized the importance of high-speed aiming and trajectory control in dual-player context.

The combination of Arduino UNO and HC-05 has enabled intuitive robot motion in several directions, with commands issued from mobile phones or joysticks [3].

L293D H-bridge motor shield is a popular choice for controlling DC motors in Arduino robots due to its reliable actuation of four-wheel steering and differential motion [4].

Solenoid control using Arduino is a well-established mechanism, with experimental reports indicating the challenges related to achieving sufficient force to propel objects with simple solenoids, highlighting the need for optimized implementation [5]. Kasaei et al. [6] developed a solenoid-based kicking mechanism for RoboCup soccer robots, which outperformed spring or pneumatic alternatives in simplicity and performance. They were able to achieve fast, variable-power strokes with compact design and rapid reload times by optimizing parameters.

There are only a few documented systems that combine Bluetooth joystick control, L293D – driven locomotion, and solenoid-based striking mechanism into a competitive dual-robot game. “PadelPro” addresses this gap by offering a novel, user-controlled interaction that integrates Arduino’s flexibility, Bluetooth input, and solenoid actuation into an innovative gameplay environment that drives skill, timing and competition.

3. Proposed Solution

The proposed solution is to develop a dual-player robotic soccer game using Arduino based robots controlled via Bluetooth joysticks. The game is played on a custom-built arena with designated goal posts, where each player maneuvers their robot and strategically activates the solenoid to strike the ball to the opponent's goal and score a point. This solution emphasizes user-controlled gameplay, modular design offering a competitive and interactive experience that brings the real-world soccer dynamics in a simple robotic form.

3.1 Analysis and Design

This microcontroller-based project integrates mechanical, electrical and wireless communication components. During the analysis phase, the functional requirements such as smooth robot movement, wireless control and ball – hitting mechanism were identified. The hardware components used include Arduino UNO for each robot, HC-05 Bluetooth module for wireless control, L293D motor driver shield to control DC motors, and a solenoid actuator to strike the ball. The design involves mapping joystick inputs to motor control signals and triggering solenoid motion. The architecture of the system follows a modular structure, separating control from actuation, ensuring ease of troubleshooting and scalability. A simple rectangular arena with goals in each side was designed to simulate a mini soccer field.

3.2 Features of Proposed Solution

PadelPro introduces a fresh idea by combining microcontroller-based robotics and remote-controlling gameplay, significantly enhancing traditional soccer games by increasing accessibility. The proposed solution is packed with various defining features such as follows;

- **Robotic Cars:** Special robot cars are provided to each player that simulate the basic movement and activities of a soccer player, enhancing the interactive element without putting any physical exertion on the players.
- **Remote-Controlled Gameplay:** Players can control their cars accurately with comfortable and ergonomic remote controllers featuring two joysticks. Smooth movements like forward, backward, left, right, and ball-striking are enabled, making the game strategic.
- **Bluetooth Wireless Communication:** Maximum car-to-remote communication is possible using Bluetooth technology with seamless and real-time response.
- **Solenoid-Powered Ball Interaction:** Each car contains a linear solenoid and a paddle, providing sufficient "kicking" motion with similar resemblance to soccer player kicking the ball.
- **Boundary Detection System:** IR sensors placed on the cars in strategic locations keep them in place inside the game court, thereby eliminating out-of-bound errors and enabling a fair game.
- **Goal Detection and Visual Feedback:** Goal posts are equipped with ultrasonic sensors that automatically detect goals, generating visual indication for clear and immediate feedback.

- **Manual Scoreboard with RFID Reset:** A manual LCD-based digital scoreboard keeps track of scores with the added advantages of reliability and convenience, and RFID technology being used to reset scores in a convenient manner.

3.3 Basic Block diagrams

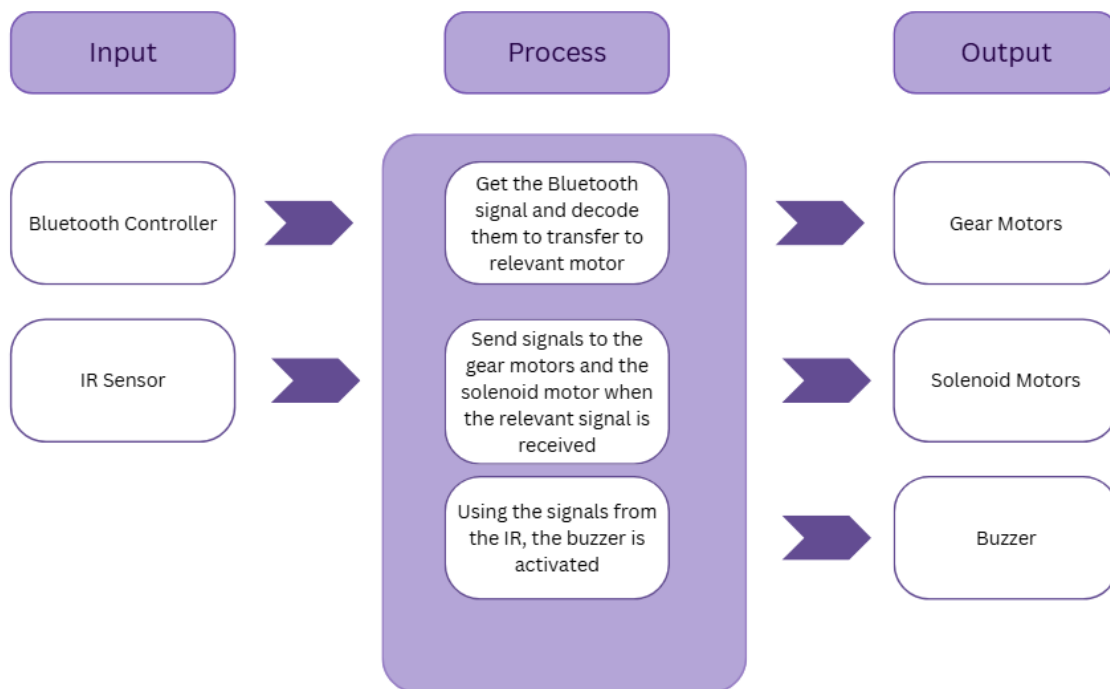


Figure 3.3.1 – Block Diagram of Receiver

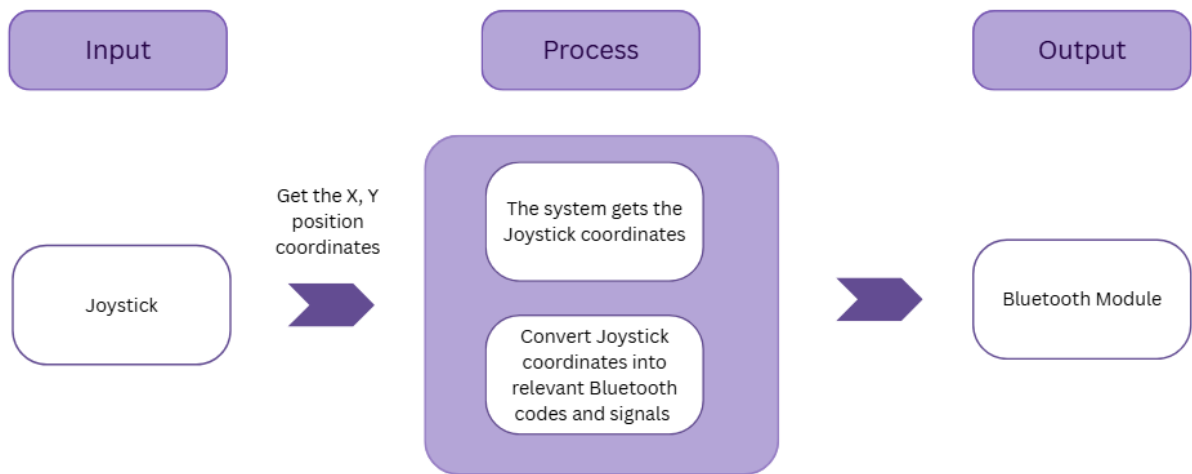


Figure 3.3.2- Block Diagram of Transmitter

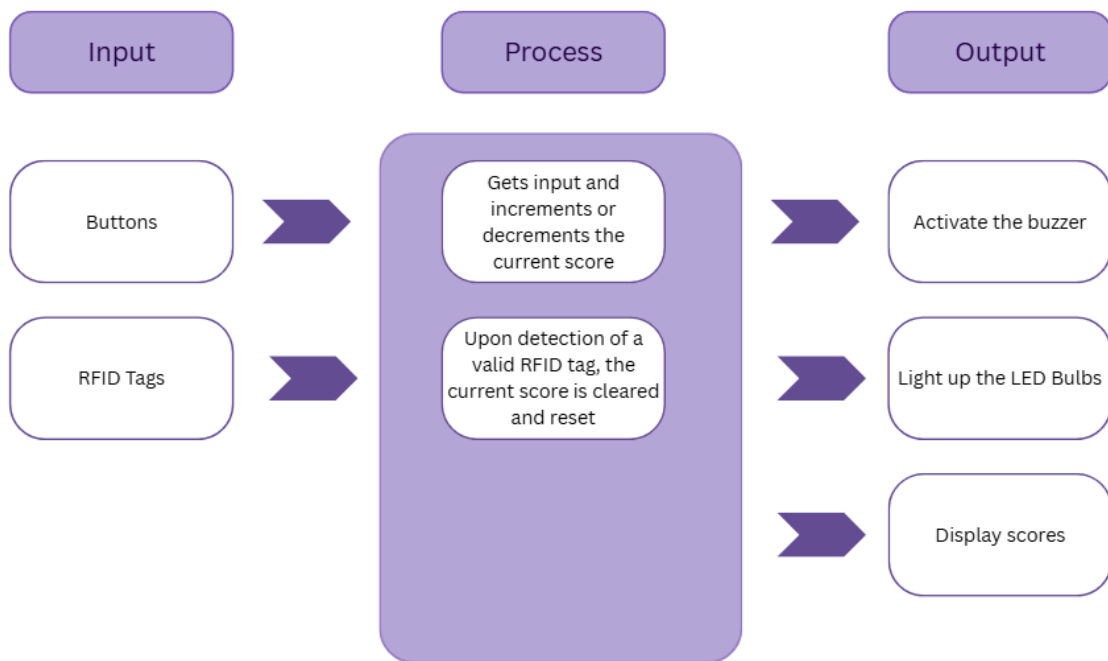


Figure 3.3.3 – Block Diagram of Scoreboard

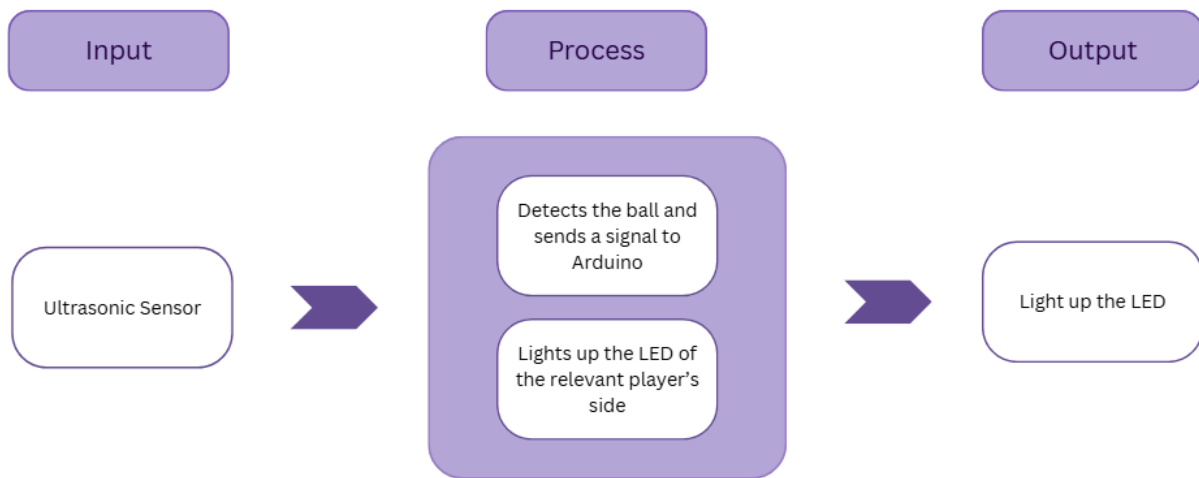


Figure 3.3.4- Block Diagram of Playground Lighting mechanism

3.4 Schematic Diagrams

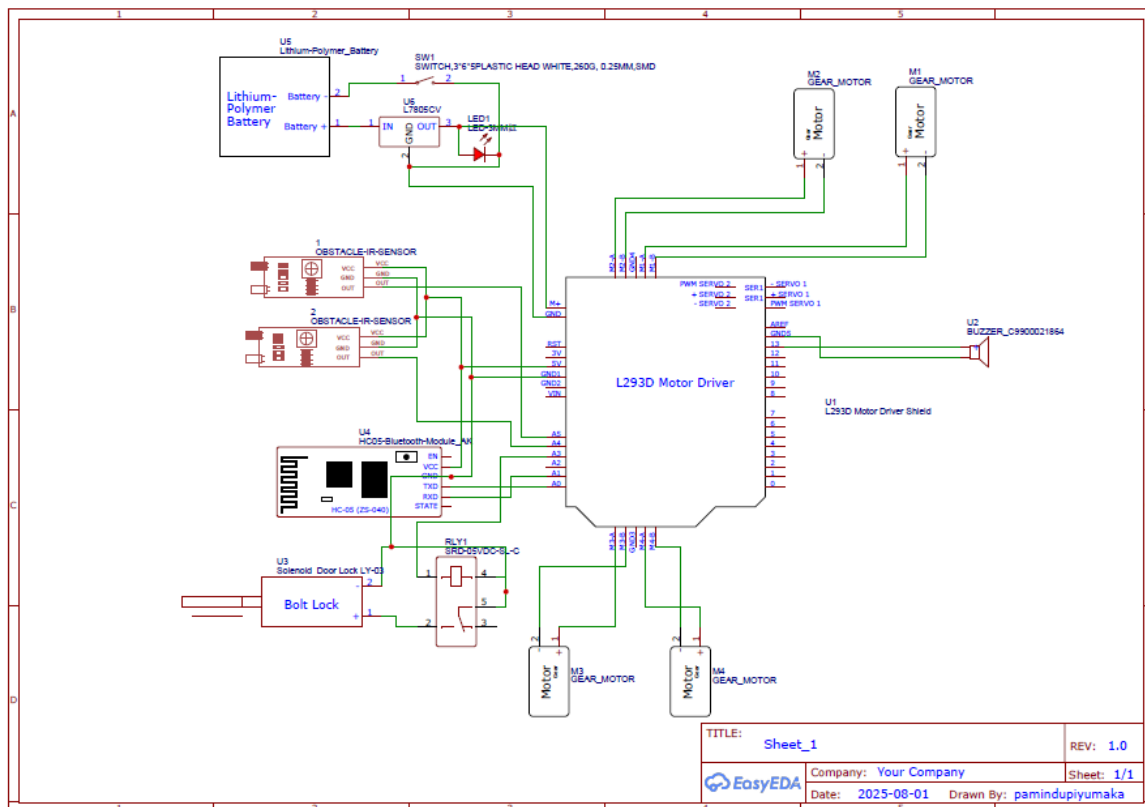


Figure 3.4.1- Receiver Circuit

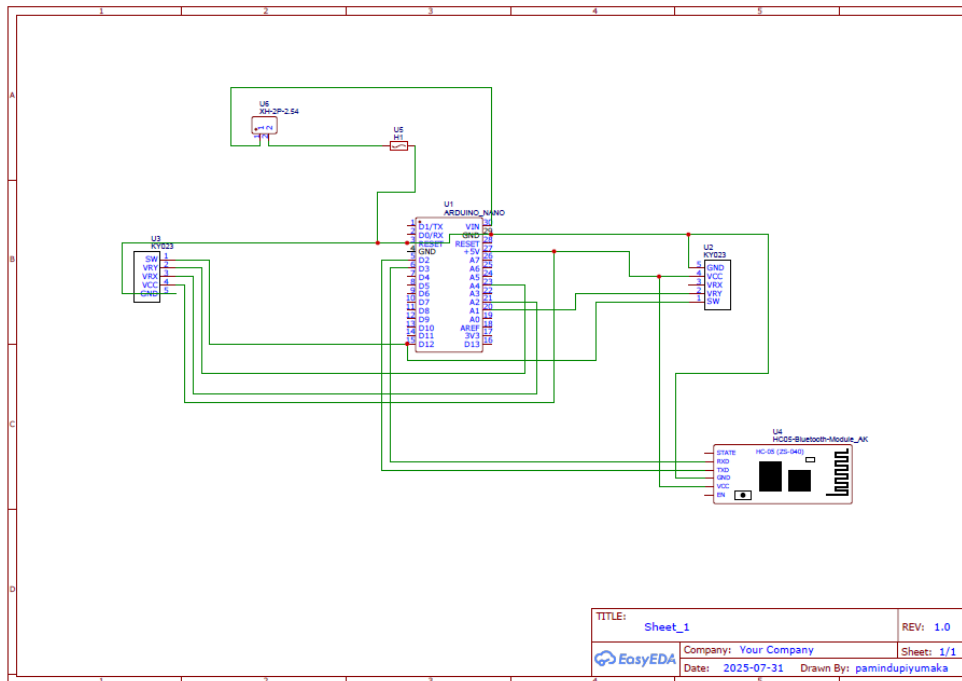


Figure 3.4.2- Transmitter Circuit

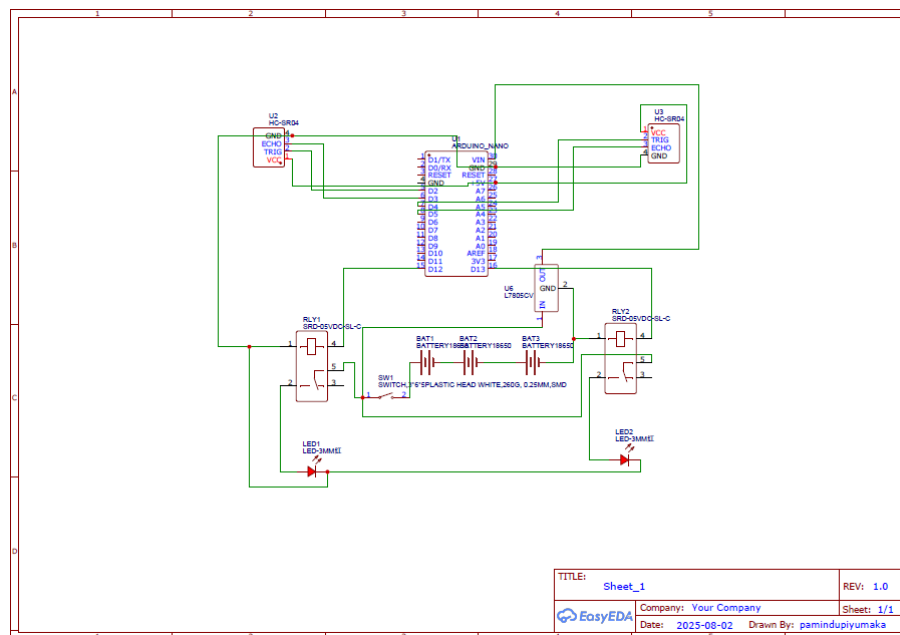


Figure 3.4.3- Playground Circuit

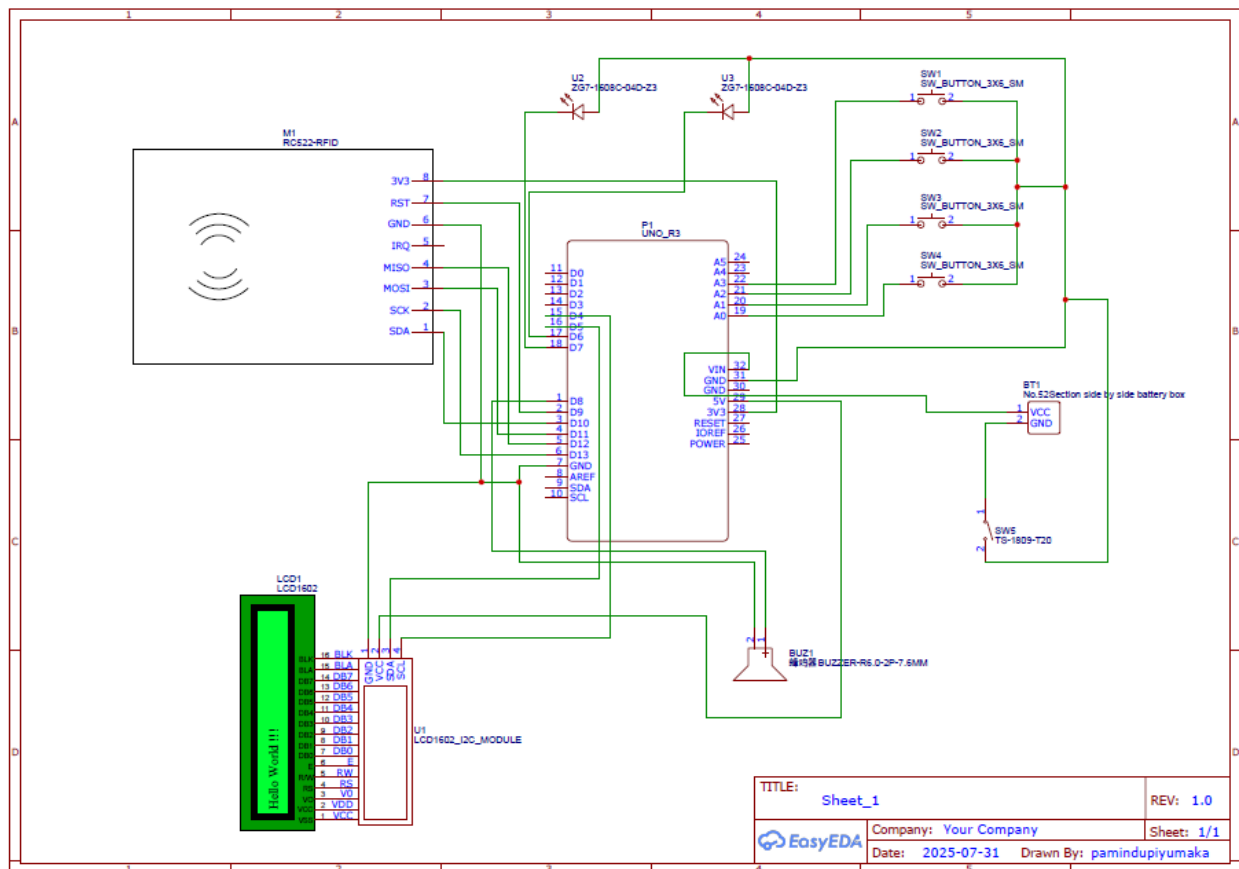


Figure 3.4.4 – Circuit Diagram of Scoreboard

3.5 PCB Design

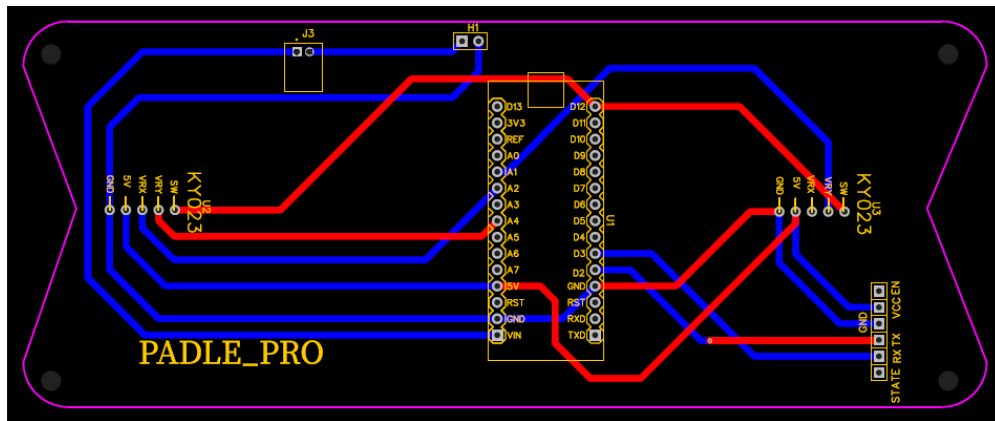


Figure 3.5.2 – PCB Design

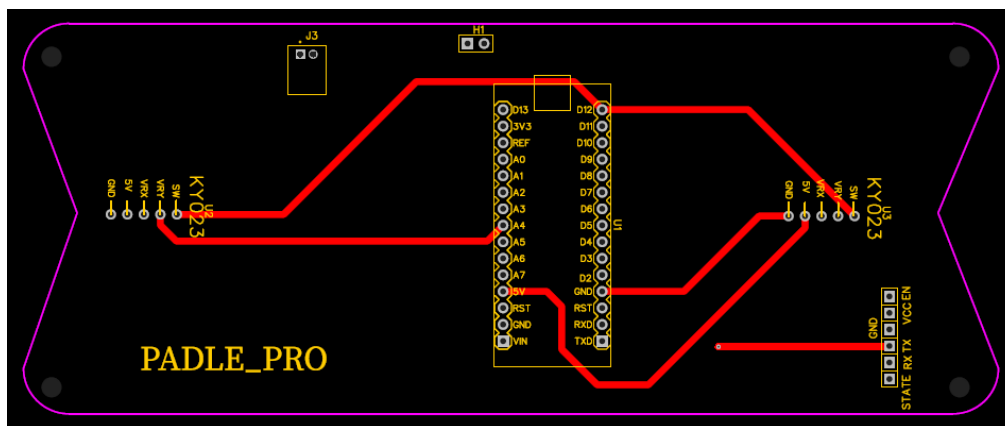


Figure 3.5.2 – Top View

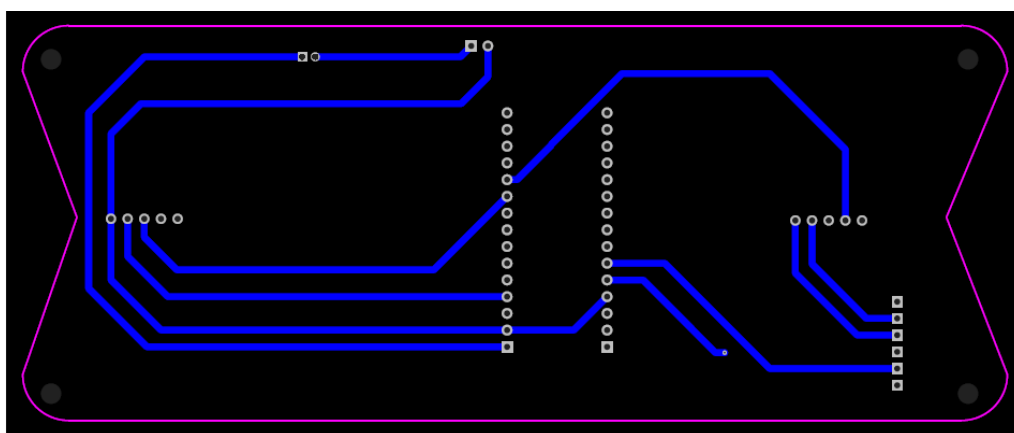


Figure 3.5.2 – Bottom View

3.4 Components

3.4.1 Arduino UNO R3



Arduino UNO is a microcontroller board based on ATmega328P. The board consists of 14 digital input/output pins, 6 analog inputs, a 16MHz ceramic resonator, USB connection, a power jack, an ICPS header and a reset button.

Pin type and function descriptions.

Pin	Function
VIN	Input Voltage (7-12V) when using external power
5V	Regulated 5V output from USB or VIN
3.3V	3.3V output
GND	Ground (0V)
RESET	Resets the microcontroller when pulled LOW
IOREF	Reference voltage for I/O pins (generally 5V)
0-13	Digital I/O pins, with pins 3,5,6,9,10,11 supporting PWM.
A0-A5	Analog inputs (0-5V), 10-bit resolution Can also be used as digital I/O

Specification	Description
Microcontroller	ATmega328P
Operating Voltage	5V
USB connector	USB-B
Power Supply connector	Barrel connector
Input Voltage (recommended) Input Voltage (limits)	7V-12V 6V-20V
Digital I/O pins	14
Analog input pins	6
DC current per I/O pin	40mA
DC current per 3.3V pin	50mA
Flash Memory	32KB(ATmega328P)
SRAM	2KB(ATmega328P)
EEPROM	1KB(ATmega328P)
Clock speed	16MHz
Dimensions	68.6mm × 53.4mm
Weight	25g

3.4.2 Arduino Nano

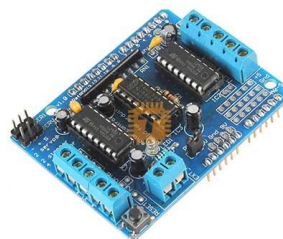
Arduino Nano is a compact, breadboard-friendly microcontroller board based on ATmega328P. The functionality of the board is similar to that of Arduino Uno, but has a much smaller footprint, making it ideal for compact or wearable devices. It consists of multiple digital and analog I/O pins, serial communication capabilities, and USB-based programming making it a popular choice in prototyping, robotics and IoT applications.



Specification	Description
Microcontroller	ATmega328P
Operating Voltage	5V
USB connector	USB-B
Power Supply connector	Barrel connector
Input Voltage (recommended) Input Voltage (limits)	7V-12V 6V-20V
Digital I/O pins	14 (6 provide PWM output)
Analog input pins	8
DC current per I/O pin	40mA
Flash Memory	32KB
SRAM	2KB
EEPROM	1KB
Clock speed	16MHz
Dimensions	18mm x 45mm
Weight	~7g

3.4.3 L293D motor drive expansion shield board

The L293D motor drive expansion shield board is an Arduino compatible board that is able to control DC motors and stepper motors. It's built around the L293D IC which is a dual H-bridge motor driver. This board enables motors to be controlled directly from the Arduino board without the need for extra components.



Specification	Description
Main IC	L293D
Motor Control	Up to 4 DC motors or 2 stepper motors
Motor Voltage	4.5V- 25 V
Logic Voltage	5V
Current per channel	600mA per channel, 1.2 A peak
PWM support	Available
Dimensions	Approximately 70 × 55 mm

3.4.4 TT Motors

A TT motor, also known as a DC geared motor is a common component used in robotic projects. Its typical operating voltage is 3V-6V and the gearbox helps reduce speed and increase torque output. The stall current can be significant, especially in high voltages, therefore appropriate motor drivers should be used.



Specification	Description
Rated Voltage	3V-6V
Continuous no-load current	150mA +/- 10%
Min. operating speed (3V)	90 +/- 10% RPM
Min. operating speed (6V)	200 +/- 10% RPM
Stall torque (3V)	0.4kg.cm
Stall torque (6V)	0.8kg.cm
Gear ratio	1:48
Body dimensions	70mm ×22mm×18 mm
Weight	30.6g

3.4.5 Omni Wheels

Omni wheels or omnidirectional wheels are a special type of wheels with small rollers mounted around their circumference. These rollers are typically placed at a 90° or 45° angle to the wheel's main axis of rotation which is a unique design that allows the robot to move forward, backward, sideways or diagonally without changing its orientation. They are commonly used in robotics competitions and mobile platforms that require a high level of maneuverability.



Specification	Description
Wheel diameter	80mm approximately
Roller angle	90° or 45
Number of rollers	6-16
Material (wheel Body)	ABS plastic
Material (Rollers)	Rubber
Load per wheel	~2–15 kg
Weight	~50–200g

3.4.6 IR Sensors

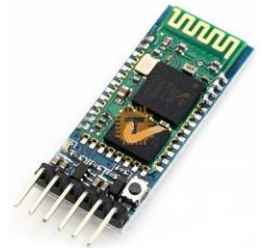
Infrared (IR) sensors are electronic sensors employed to detect objects or distance by emitting infrared radiation and detecting its reflection off an obstacle. They consist of an infrared transmitter and receiver working together for proximity detection. IR sensors are applied in robotics, automatic machinery, obstacle avoidance systems, and line-following robots due to their small size, accuracy, and fast response.



Specification	Description
Detection range	Typically, adjustable between 2cm to 30cm
Operating Voltage	3.3V to 5V DC.
Number of rollers	6-16
Output type	Digital
Response speed	High, suitable for real-time applications
Current consumption	20mA – 40mA
Wavelength	~940 nm (IR LED)
Sensor Type	Reflective (IR LED + Photodiode/Phototransistor)

3.4.7 HC-05 Bluetooth Module

HC-05 Bluetooth module is an efficient wireless communication module that allows serial data transmission and reception between electronic devices. It leverages the Bluetooth V2.0 + EDR protocol in the 2.4 GHz ISM frequency band in order to offer a dependable way to create short-distance wireless connections.



Specification	Description
Bluetooth Version	2.0 + EDR (Enhanced Data Rate)
Operating Voltage	3.3V – 6V
Operating Current	~30 mA
Communication Interface	UART (TX, RX)
Baud Rate (Default)	9600 bps
Range	~10 meters
Modes	Master / Slave

3.4.8 Joysticks

Thumb joystick modules are often used in gaming projects as input devices that offer user-friendly direction control. These modules are equipped with two orthogonal potentiometers used to produce analogue voltage signals proportional to their position in order to detect movement in the X and Y axes. These joysticks normally consist of a push-button switch that is pressed by the joystick shaft.



Specification	Description
Operating Voltage	5V
Operating Voltage	3.3V – 6V
Analog Output	Two axes (X, Y) with neutral centering
Digital Output	Integrated press-to-click button
Typical Current Draw	~10mA
Dimensions	~34 x 26 x 20 mm
Movement Range	~ $\pm 10^\circ$ from center on each axis

3.4.9 6V Push Type Solenoid (10mm Stroke)

A solenoid is an electromagnetic actuator that converts electrical energy to linear mechanical motion. They have a coil of wire around a moving metal core or plunger. The coil is energized by an electric current which propels the plunger. Linear push-pull solenoids are a convenient solution to mechanisms that require fast, precise, and controlled movements.



Specification	Description
Rated Voltage	6V
Current	~0.8 A to 1.2 A
Power Consumption	~5W to 7W
Stroke Length	10mm
Force (at rated voltage)	0.5 – 1.5 kgf
Plunger Return	Spring-loaded return when de-energized
Body Material	Metal housing
Dimensions	~28mm × 15mm × 13mm

3.4.10 16 ×2 LCD display

A 16 ×2 LCD display is a type of alphanumeric display that can show 2 lines of 16 characters each. They are commonly used in microcontroller-based projects to display text outputs.



Specification	Description
Display format	16 characters × 2 lines
Character size	Typically, 5×8 dot matrix
Operating Voltage	4.7V- 5.3V (commonly 5V)
Interface Type	Parallel
Backlight	LED
Dimensions	80mm ×36mm ×12mm

3.4.11 RFID sensor

A low frequency RFID sensor, typically operating at 125kHz is a contactless technology used for reading RFID cards or tags within a distance of approximately 10cm. They are inexpensive, reliable and easily integrated with microcontrollers like Arduino.



Specification	Description
Operating frequency	125kHz
Reading distance	2-10cm
Operating Voltage	3.3V- 5V
Antenna	Built-in coil
Power consumption	Low (usually under 50mA)
Baud rate	9600bps

3.4.12 I2C Module

An I2C (Inter-Integrated Circuit) module is a small board that simplifies the connection of an LCD (usually 16x2 or 20x4) to a microcontroller like Arduino. Instead of using 6–8 digital pins, the I2C module allows communication using only two wires: SDA (data line) and SCL (clock line). This greatly reduces the pin usage and wiring complexity, especially in projects where there is a lack of I/O availability.



Specification	Description
Operating Voltage	5V
Compatible LCDs	16x2, 20x4
Backlight Control	On/off jumper or programmable via code
Contrast Adjustment	Built-in potentiometer

3.4.13 Piezo Buzzer

A Piezo Buzzer is a small, light-weighted, and energy-efficient device that uses piezoelectric effect to produce sound (beeps or tones). It is a passive buzzer commonly used in Arduino projects.



Specification	Description
Operating Voltage	3V -5V
Typical sound level	~ 85dB at 10cm
Resonant Frequency	~2 kHz – 3 kHz
Frequency range	1kHz – 8kHz
Current consumption	~5mA – 30mA
Size	~12mm diameter

3.4.14 HC SR04 Ultrasonic Sensor

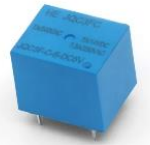
HC-SR04 ultrasonic sensor is used to estimate distance by sending high-frequency sound waves and detecting the returning echo from objects within its reach. This electronic sensor consists of two components. One is an ultrasonic transmitter which transmits the sound pulses, and the other is a receiver that detects the incoming echo. The sensor first calculates the amount of time it takes for the echo to return and gives a precise and accurate distance reading.



Specification	Description
Operating Voltage	5V DC
Detection range	~2cm to 400cm
Accuracy	± 3mm
Measuring Angle	15 degrees
Trigger Input Signal	10μs TTL pulse
Output Signal	Echo pulse proportional to distance

3.4.15 5V Relay

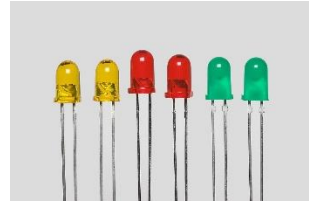
A 5V relay module is an electromechanical switch used to control high-voltage devices with a low-voltage signal, such as those provided by microcontrollers like Arduino. When a 5V signal is applied to the relay input, it energizes an internal coil that closes (or opens) the switch, allowing current to flow through a separate circuit. This makes it ideal for safely switching appliances, lights, motors, and other high-voltage loads in embedded systems.



Specification	Description
Operating Voltage	5V DC
Trigger Voltage	3.3V – 5V
Maximum Switching Voltage	250V AC / 30V DC
Maximum Switching current	10A
Control signal type	Digital
Input pins	VCC, GND, IN

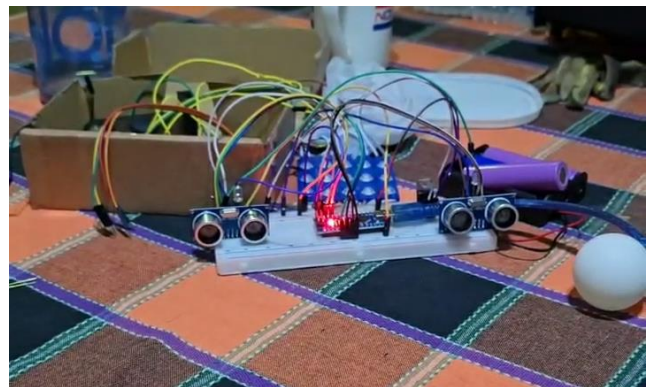
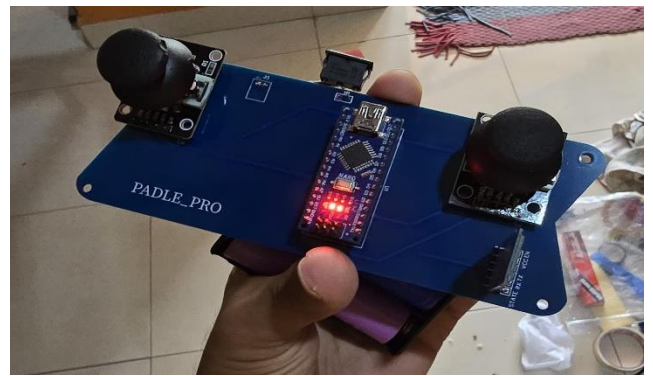
3.4.16 LED Lights

Light Emitting Diode or LED is a semiconductor device that emits light when current passes through it. LEDs are popularly used in electronic devices for indication, status indication, lighting, as well as visual effects. LEDs are energy efficient, long-lasting, and available in various color options as well as various sizes.



Specification	Description
Operating Voltage	~2V (Red, Green, Yellow), ~3V (Blue, White)
Forward Current	10 – 20 mA
Power Consumption	~40 – 60 mW
Size (Diameter)	Commonly 3mm, 5mm, and 10mm
Polarity	Anode (+) and Cathode (–) leads
Lifetime	Typically, 25,000 to 50,000 hours

4. Testing and Implementation



5. Further Work

In order to enhance the technical sophistication, user experience and functionality, the following improvements could be done as a part of future development.

- Implementation of an automatic scoring system
Incorporate a goal detection mechanism to detect and register goals and communicate the detections via Bluetooth to a central display unit or web application, eliminating the need for manual score keeping.
- Transition to an autonomous opponent robot:
Convert one of the player-controlled robots to an autonomous unit capable of ball tracking and executing movements using onboard sensors. This would introduce Artificial Intelligence elements to the game and transform it to single player mode.
- Integration of advanced sensors for improved gameplay dynamics:
The robots can be equipped with proximity sensors or gyroscopes for additional features such as collision avoidance, automatic orientation correction and semi-autonomous behavior.
- Introduction of additional game modes:
The game's functionality can be expanded by introducing timed matches, score-based limits or interactive obstacles to improve enjoyability and interactivity.

6.Budget

Component	Quantity	Price (LKR)
Arduino UNO board	2	2750
Arduino NANO board	3	1,200
L293D motor drive expansion shield board	2	470
TT motors	4	200
Omni wheel sets	2	1,600
Relay	4	170
Solenoid Switch	2	1260
Arduino type joysticks	4	250
Li- Poly RC batteries	2	3,200
18650 RC Batteries	7	340
Buzzer	3	95
RFID tag and reader	1	210
IR sensors	2	150
HC-05 Bluetooth module	4	1,000
LCD screen	1	320
Push buttons	4	20
Battery casing	3	50
UNO cable	2	175
PCB		8155
Acrylic and Plywood Boards		10760

Laser cutting		10380
Ultrasonic sensor	2	300
LED strips	10	50
Miscellaneous Expenses		10,000
Total		73110

7. Workload Matrix

W.M.C.S. Wijekoon (234232B)	Develop the soccer robot. Integrate the code for the robot. Develop the scoreboard control system.
W.P.P. Welikada (234224D)	Develop the soccer robot. Integrate the code for the robot. PCB design.
H. M. Kularatne (234116A)	Designing of the chassis and scoreboard. Implementing the code for the Bluetooth Joystick. Develop the playground.
S. S. Ratnayake (234182U)	Develop the Bluetooth Joystick. Implementing the code for the Bluetooth Joystick. Develop the game logic.
K. K. G. Saputhanthri (234189X)	Designing the joystick and playground. Implementing the playground control system. Development of the scoreboard.

7.1 W.M.C.S. Wijekoon (234232B)

7.1.2 Development of the soccer robot

To lead the development of a functional robotic car to be used as a soccer robot in the gaming system. Responsible for assembling the mechanical structure by integrating essential parts such as motors, wheels, sensors and chassis. The task involved detailed and precise attention paid to component placement, weight distribution, and wiring to support smooth movement and real-time control. The completed car served as a key component in the gaming system.

7.1.2 Developed the embedded software for car operation

Programmed the embedded system controlling the robotic car, focusing on accurate motion control, response to external inputs and seamless communication with other system components by facilitating operations such as motor control, directional movement, and event-triggered actions. Developed primarily for Arduino UNO microcontroller, this ensures the car's responsiveness and smooth performance during gameplay.

7.1.3 Development of the scoreboard control system

Responsible for developing the embedded code that controls the scoreboard. The code handles input detection from buttons, manage score variables in memory and update display in real time to display current score. Emphasis was placed on responsiveness and ease-of-use allowing the scoreboard to function accurately without delays.

7.2 W.P.P. Welikada (234224D)

7.2.1 Development of the Soccer Robot

Contributed to the hardware development of the soccer robot by particularly focusing on the solenoid kicking mechanism, Bluetooth connectivity, and sensor interfacing, thereby implementing game specific activities such as kicking the ball, providing directional inputs

from the remote to the car, and getting a sound output when the car crosses boundary lines of the playground by using a combination of IR and buzzer system.

7.2.2 Designed the PCB for the Remote controller

Responsible for designing a custom PCB for the handheld Bluetooth remote controller. This involved the schematic design, component selection, and PCB layout using “EasyEDA” design tool. The PCB layout was designed to accommodate joysticks, Bluetooth transmission module, power supply components, and microcontroller connections in a compact and efficient format.

7.2.3 Developed the embedded code for car operation

Developed the code that controlled the robotic car’s Bluetooth communication, kicking action and directional movement. The code was developed to process incoming signals from the remote and determine the appropriate motor actions and trigger the solenoid actuator. Additionally, the code handles input from IR sensors to assist navigation and provide a sound output through a buzzer when boundary lines are crossed. This ensured that the car is accurate in its motion, capable of both movement and interaction functions, and is responsive.

7.3 H.M. Kularatne (234116A)

7.3.1 Designing of Chassis and Scoreboard Layout.

Responsible for the physical design of the chassis which included careful consideration of the weight distribution, component placement and accessibility, ensuring the system has a functional and stable base to integrate electronic components and implement gameplay movements. In addition, the physical layout of the scoreboard was also designed, creating a clear and compact structure that housed the LCD display, input buttons and RFID sensor.

7.3.2 Development of Bluetooth remote code

Developed the embedded code for the Bluetooth remote controller, focusing mainly on establishing and managing master slave communication between the remote and the robotic car. The remote (master) was programmed to initiate and control the Bluetooth connection with the car (slave), ensuring reliable transmission of directional commands and kicking signals. The code was developed to enable real-time control and maintain stable communication throughout gameplay, enhancing the user experience.

7.3.3 Development of the Playground

Responsible for the development of the interactive playground which visually responds when a goal is scored. The system uses LEDs placed on the goal posts, which are programmed to light up when the ball enters the goal area. This was achieved by using Ultrasonic sensors and programming the LEDs to respond based on trigger events. This feature contributed to both the aesthetic and functional aspects of the overall system.

7.4 S.S. Ratnayake (234182U)

7.4.1 Developed the Bluetooth remote controller

Responsible for the development of the Bluetooth remote controller which serves as the primary user interface for controlling the robotic car. The task involved selecting appropriate components such as Bluetooth module, joysticks, and voltage supply and integrating them into a functional handheld device. Emphasis was placed on making the final product compact and easy to use while also having a reliable power management, making it suitable for real-time gameplay.

7.4.2 Development of embedded code for the remote controller

In parallel to the hardware development, the embedded code for the remote was also developed. Established robust communication between the car and the remote by handling joystick input scanning, data formatting, and transmitting via the Bluetooth module., ensuring timely communication. The implementation prioritized reducing latency, stability and responsiveness, all of which are essential for smooth gaming experience.

7.4.3 Development of game logic

Defined and implemented the overall game logic that governs how each component interacts with the other within a game session. This included setting rules for scoring, turn taking, and winning conditions. The logic was designed to ensure a balanced, repeatable and engaging gameplay experience. This serves as the foundation upon which all hardware and software components were developed.

7.5 K.K.G. Saputhanthri (234189X)

7.5.1 Designed the remote and playground layout

Responsible for the designing the physical layout of both the remote controller and playground. The remote was designed with emphasis to user comfort and ease of use, ensuring intuitive operation. The playground design involved strategic placement of goal posts, boundary lines and sensor zones by ensuring ample play area is left for the user to move the robot around. These designs helped build a consistent and engaging gaming environment.

7.5.2 Developed the code for the playground

Developed the embedded software controlling the goal detection lighting mechanism. When a goal is detected using ultrasonic sensors during gameplay, the code activates the LED indicators around the goal post, providing visual feedback to the players and spectators. This mechanism makes the experience more immersing and engaging.

7.5.3 Development of the Scoreboard

Developed the digital scoreboard system to update player scores in real time manually by a button press. The system also incorporates a RFID sensor to establish start of game and reset scores. A 16 x 2 LCD display is used to display player score details. The scoreboard provides a critical interface for tracking game progress and providing clarity in a competitive gaming environment.

9. References

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